

Prayas JEE 2025

Maths

DPP: 1

Trigonometric Functions

Q1 $\sin\left(\frac{17\pi}{3}\right)$ is equal to

- (A) $-\frac{1}{2}$
 (B) $\frac{1}{2}$
 (C) $\frac{\sqrt{3}}{2}$
 (D) $-\frac{\sqrt{3}}{2}$

Q2 $\tan\left(\frac{11\pi}{3}\right)$ is equal to:

- (A) $\sqrt{3}$
 (B) $\frac{1}{\sqrt{3}}$
 (C) $-\frac{1}{\sqrt{3}}$
 (D) $-\sqrt{3}$

Q3 What are the values of $\cos A - \sin A$, when value of A is $\frac{2\pi}{3}$

- (A) $\left(\frac{\sqrt{3}-1}{2}\right)$
 (B) $\left(\frac{1-\sqrt{3}}{2}\right)$
 (C) $\left(\frac{1+\sqrt{3}}{2}\right)$
 (D) $-\left(\frac{1+\sqrt{3}}{2}\right)$

Q4 What are the values of $\tan A + \cot A$, when value of A is $\frac{7\pi}{4}$

- (A) 1
 (B) 0
 (C) -1
 (D) -2

Q5 Which of the following is 0 at $x = n\pi \in I$

- (A) $\tan x$
 (B) $\sin x$
 (C) $\cot x$
 (D) $\cos x$

Q6 Which of the following is / are correct, when $n \in I$

- (A) $\sin(2n+1)\pi/2 = 1$
 (B) $\cos(2n+1)\pi = 1$

(C) $\sin(4n+1)\pi/2 = 1$

(D) $\cos(2n\pi) = 1$

Q7 Which of the following is / are correct, when $n \in I$

- (A) $\tan(2n+1)\pi/2$ is not defined
 (B) $\sec(2n+1)\pi/2$ is not defined
 (C) $\operatorname{cosec} n\pi$ is not defined
 (D) $\cot n\pi$ is not defined

Q8 Which of the following is/are correct?

- (A) $4053\pi/2$ lies on boundary of first & second quadrant
 (B) $4053\pi/2$ lies on boundary of third & fourth quadrant
 (C) -7043° lies in quad II
 (D) 7043° lies in quad III

Q9 Which of the following is/are correct?

- (A) $\cos(502\pi) = 1$
 (B) $\sin(4053\pi/2) = 1$
 (C) $\cos(507\pi) = -1$
 (D) $\tan 7\pi = 0$

Q10 Which of the following is / are correct

- (A) $\sin(100\pi) = 0$
 (B) $\sin(399\pi/2) = -1$
 (C) $\cot 5\pi/2 = 0$
 (D) $\tan 7\pi/2$ is not defined

Q11 Find quadrant in which following angles lie

- (i) 765°
 (ii) 5253°
 (iii) -8054°



(iv) 2030°
(v) -7156°



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Answer Key

Q1 (D)

Q2 (D)

Q3 (D)

Q4 (D)

Q5 (A, B)

Q6 (C, D)

Q7 (A, B, C, D)

Q8 (A, C, D)

Q9 (A, B, C, D)

Q10 (A, B, C, D)

Q11 i) First quadrant
ii) Third quadrant
iii) Third quadrant
iv) Third quadrant
v) First quadrant



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